CTS — Aptitude Test 1

Questions 1–25 of 100 · 25 questions

Question 1. If log (p+q)= -1, then what is the value of log(p+q)(p2– q2)?

Solution:
log(p+q)(p2 – q2)
= log(p+q)(p+q)(p – q)
= log(p+q)2(p – q)
= log(-1)2(p – q)
= log(1)(p – q)
= 0 + log(p – q)
= log(p – q)

Question 2. log4 2+log4 32 is equal to?

Solution:
log4 2 + log 4 32 = log4 (2x32)
= log4 64
= log4 43

= 3 log4 4
= 3 x 1 = 3

Question 3. Find the value of log1+log2+log3?

Solution:
log 1 = 0
log 2 = 0.3010
log 3 = 0.4771
log 1 + log 2 + log 3 = 0.7781

Question 4. Find the missing number. 3 , 11 , 25 , 45 ,__

Solution:
11 – 3 = 8
25 – 11 = 14
45 – 25 = 20
The difference between the successive terms is an arithmetic progression with common difference 6.
So, the next difference will be
20 + 6 = 26
The missing number is
45 + 26 = 71

Question 5. Find the missing number in the series 1, 11, 21, 1211, 111221, ____.

Solution:
The second term in the series represents that there is ‘one’ 1 i.e. ‘1’1 = 11

The third term represents that there are
‘two’ 1’s in the second term i.e. ‘2’1 = 21
The fourth term represents that there is
‘one’ 2 and ‘one’ 1 i.e. ‘1’2 ‘1’1 = 1211
So, the missing term is ‘three’ 1 ‘two’ 2
‘one’1 i.e. 312211

Question 6. Find the missing term 16, 136, 1096, ?

Solution:
(16 x 8) + 8 = 136
(136 x 8) + 8 = 1096
Therefore the missing number is,
(1096 x 8) + 8 = 8776

Question 7. Find the missing number 25,168, ? , 8176

Solution:
25 x 7 – 7 = 168
168 x 7 – 7 = 1169
1169 x 7 – 7 = 8176
Therefore the missing number is,
1169

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Question 8. Find the missing number 24,39,416,525,___

Solution:
24 = 2, 4 = 4 is the square of 2
39 = 3, 9 = 9 is the square of 3
416 = 4, 16 = 16 is the square of 4
525 = 5, 25 = 25 is the square of 5
Therefore the missing number is,
6, square of 6 = 6, 36
= 636

Question 9. Find the missing number 8,12,24,60,___

Solution:
8 x 1.5 = 12
12 x 2 = 24
24 x 2.5 = 60
60 x 3 = 180
Missing number is 180

Question 10. Find the missing number 10,7,12,10,14___

Solution:

10 – 3 = 7
7 + 5 = 12
12 – 2 = 10
10 + 4 = 14

The difference between the terms is
decreasing
and
increasing
in
the
following manner
-3, +5, -2, +4
The next change will be -1.
The missing term is 14 – 1 = 13

Question 11. Find the missing number 3,6,9,_,24,36.

Question 12. Find the missing term 3, 6, 12, 48, 29, ___

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Solution:
Take the numbers as pairs.
(3,6), (12,48), (29, x)
(3,6)
= 3, 3x21

Question 13. If 5+4 = 2091; 6+3 = 1893; 7+5 = 35122; then 9+8 = ?

22058 = 2(1+37+n)

So,
2058 = 1+37+n
n = 2058 – 38
n = 2020

Solution:
5+4 = 2091
First two digits of 2091 = 20 = 5 x 4
Third digit of 2091 = 9 = 5 + 4
Last digit of 2091 = 1 = 5 – 4
Therefore,
For 9 + 8
First two digits = 9 x 8 = 72
Next two digits = 9 + 8 = 17
Last digit = 9 – 8 = 1
The value of 9 + 8
= 72171

Question 14. What is the remainder when we divide 125! by 1031?

Solution:
The number of zeros at the end of 125!
= (125/5) + (125/25) + (125/125)
= 31
The number of zeros at the end of 1031

=31
So, when 125! is divided by 1031, the
reminder = 0

Question 15. If all 6's are replaced by 9, the algebraic sum of all numbers from 1 to 100 (both inclusive)varies by?

Solution:
The sixth term is 4 x (third term)
4 x 9 = 36
The fifth term is 4 x (second term)
4 x 6 = 24
Therefore the fourth term
= 4 x (first term) = 4 x 3
= 12

Solution:
Total number of 6's in units place from 1 to 100 = 10
Increase in sum by changing all the 6's in units place to 9 will be = 10x3 = 30
Total number of 6's in tens place from 1 to 100 = 10.
Increase in sum by changing all the 6's in tens place to 9 will be = 10x30 = 300
Total increase = 300 + 30 = 330

Question 16. 274+22058+22n . For what value of n the expression is a perfect square?

(12, 48)
= 12, 12x22

Solution:
(a+b)2 = a2 + 2ab + b2

(29, x)
= 29, 29x23

Consider a2 = 274 ; b2 = 22n

The missing number is,
29 x 23 = 29 x 8 = 232

So, a = 237 ; b = 2n

For the expression to be a perfect square
22058 = 2 x 237 x 2n

Question 17. The value of 6(-2) is?

Solution:
6(-2) =1/(62)
= 1/36

Question 18. If X^Y denotes X raised to the power of Y, find

out last two digits of (2957^3661) + (3081^3643).

Solution:
Last two digits of 29573661

The base ends in 7.
573661 = 574(915) x 571

(572 x 572)915 x 57
Last two digits of 572 = 49
So,
(49x49)915 x 57
Last two digits of 49x49 = 01
So, last two digits of 29573661

= (01)915 x 57 = 57
Last two digits of 30813643

Last digit will be 1.
Second last digit(only if the base ends in 1) = unit place of (tens digit of base x unit digit of the power value)
= unit place of (8 x 3)
= unit place of 24 = 4
Last two digits of 30813643 = 41
Last two digits of the expression
= 57 + 41
= 98

Question 19. The reciprocal of the HCF and LCM of two numbers are 1/12 and 1/312. If one of the number is 24 then the other number is:

Solution:
The HCF and LCM are 12 and 312 respectively.
Product of two numbers = product of their HCF and LCM
24 x N = 12 x 312
N = 156
The second number = 156

Question 20. Sum of the digits of a 3-digit number is subtracted from the number. The resulting number is always divisible by?

Solution:
Let the three digit number be xyz.
Sum of the digits = x+y+z
xyz can be written as 100x + 10y + z
100x + 10y + z – x – y – z = 99x + 9y

After subtracting the sum of the digits from the actual number, the resulting number will always be a multiple of 9.

Question 21. More than half of the members of a club are ladies. If 4/7 of the ladies and 7/11 of the gents in the club attended the meeting, then what is the smallest number that club could have?

Solution:
According to the fractions given in the question, if the number of ladies = 7 and number of gents = 11, the conditions given in the question are not satisfied.
So, double the fraction of number of ladies in the club.
New fraction = 8/14
Now we have 14 ladies and 11 gents, which satisfy the condition.
Minimum number of members in the club = 14 + 11 = 25

Question 22. If mini download three more songs in her mobile, she will have songs with 512MB in her mobile. If on an average each song is 4 MB, how many songs did she initially have in her phone before downloading?

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Solution:
If each song is 4 MB, the total MB before downloading the three songs
= 512 – (3x4)
= 500 MB
Number of songs she had initially
= 500/4 = 125 songs

Question 23. What is the unit digit of the following expressions: 1+2^2+3^3+4^4+5^5+.....10^10

Solution:
Unit place of 11 = 1
Unit place of 22 = 4
Unit place of 33 = 7
Unit place of 44 = 6
Unit pace of 55 = 5
Unit place of 66 = 6
Unit place of 77 = 3

Unit place of 88 = 6
Unit place of 99 = 9
Unit place of 1010 = 0
Unit place of the expression

= 1+4+7+6+5+6+3+6+9+0
= unit place of 47
= 7

The first number in the series which is divisible by 2 = 100.
The last number that is divisible by 2 = 200
Common difference between each terms that are divisible by 2 = 2
Using the formula to find out the number of terms in an arithmetic series:
N = [(l – a)/d] + 1
Number of terms divisible by 2
= [(200 – 100)/2] + 1
= 51
The first number in the series that is divisible by 3 = 102
The last number that is divisible by 3 = 198
Number of terms = [(198-102)/3]+1
= 33
In both the series, e have numbers which are both a multiple of 2 and a multiple of 3, that has to be removed.
The numbers which are divisible by 2 and 3 are divisible by 6.
The first term divisible by 6 = 102
The last term divisible by 6 = 198
Number of terms = [(198-102)/6]+1
= 17
Number of terms which are either divisible by 2 or 3
= 51 + 33 – 17
= 67

Question 24. Divide the sum of 3/5 and 8/11 by their difference.

Solution:
Sum of 3/5 and 8/11
= 3/5 + 8/11 = (33+40)/55
= 73/55
Difference between 3/5 and 8/11
= 8/11 – 3/5
= (40 – 33)/55
= 7/55
Divide 73/55 by 7/55
We get
(73/55)/(7/55)
= 73/7

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Question 25. What is the smallest number ,which when divided by 7,18,56 and 36, leaves a remainder 0?

Solution:
The smallest number is the LCM of the given numbers.
LCM of (7, 18, 56, 36)
= 7584